

Partial Derivatives

a Partial derivative does not just have one but multiple variables, a regular derivative would just have a x but with partial derivatives you get a x, y, z and you will find the derivative of one while you treat the rest as constants

★ "∂" is not the letter d but that is a symbol for partial derivatives

Notation for the first partial derivative: $\frac{\partial F}{\partial x} = F_x$

Notation for second partial derivative: $\frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} \right) = F_{xy}$

$$F(x, y) = (y^2 \ln x + 7x - 4y)^{10}$$

$$\frac{\partial F}{\partial x} = F_x = 10(y^2 \ln x + 7x - 4y)^9 \cdot \frac{y^2}{x} + 7$$

• Partial derivative taken with respect to "x"

$$\frac{\partial F}{\partial y} = F_y = 10(y^2 \ln x + 7x - 4y)^9 \cdot 2y \ln x - 4y$$

• Partial derivative taken with respect to "y"

$$F(x, y) = 7x^3 - 4y^5 + 10x \sin y$$

$$\frac{\partial F}{\partial x} = 21x^2 - 0 + 10 \sin y$$

$$\frac{\partial F}{\partial y} = -20y^4 + 10x \cos(y)$$

$$\begin{array}{ccc}
 f_{xx} & , & f_{yx} & , & f_{yxy} \\
 \downarrow & & \downarrow & & \downarrow \\
 \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) & & \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) & & \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \right)
 \end{array}$$

these are First, second and third Partial derivatives but each is asking for a different variable you always began with the First given letter

$$\frac{\partial f}{\partial x} = 21x^2 + \sin(y)$$

• here you are already given the $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

$$\frac{\partial^2 f}{\partial x^2} = f_{xx} = 42x$$

$$\frac{\partial f}{\partial y} = -20y^4 + 10x \cos(y)$$

$$\frac{\partial^2 f}{\partial x \partial y} = f_{yx} = 0 + 10 \cos(y)$$

$$f(x, y, z) = 10x^4 + 7y^2 - 2 \cos(x)$$

Find f_{yxy}

$$\frac{\partial f}{\partial y} = 0 + 14y - 0 = 14y$$

• so the final answer is just zero

$$\frac{\partial f}{\partial x \partial y} = 14y = 0$$

$$\frac{\partial f}{\partial x \partial y^2} = 0 = 0$$

Find the indicated partial derivative of $f(x, y, z) = x^{yz^2}$

$$f_x(e, 1, 0)$$

$$f = x^{yz^2}$$

$$\ln(f) = yz^2 \ln(x)$$

$$\frac{1}{f} \cdot \frac{\partial f}{\partial x} = yz^2 \cdot \frac{1}{x}$$

$$\frac{\partial f}{\partial x} = f \cdot \frac{yz^2}{x}$$

• Plug x^{yz^2} into f

$$\frac{\partial f}{\partial x} = x^{yz^2} \cdot \frac{yz^2}{x}$$

$f_x(e, 1, 0)$ Plug these into the letters
 $x \ y \ z$

$$f_y(e, 1, 0)$$

$$e^{(1)(0)^2} \cdot \frac{(1)(0)^2}{e} = \boxed{0}$$

$$f = x^{yz^2}$$

$$\ln(f) = yz^2 \ln(x)$$

$$\frac{1}{f} \cdot \frac{\partial f}{\partial x} = z^2 \ln(x)$$

$$\frac{\partial f}{\partial x} = f \cdot z^2 \ln(x)$$

$$\frac{\partial f}{\partial x} = (x^{yz^2}) z^2 \ln(x) = x^{yz^2} z^2 \ln(x)$$

Given $xz + x \ln y = z^2$

Find $\frac{\partial z}{\partial x}$

$$xz + x \ln y = z^2$$

$$\frac{\partial f}{\partial x} \cdot g + f \cdot \frac{\partial g}{\partial x}$$

$$f = x \quad g = z$$

$$\frac{\partial f}{\partial x} = 1 \quad \frac{\partial g}{\partial x} = \frac{\partial z}{\partial x}$$

$$(1)(z) + (x) \left(\frac{\partial z}{\partial x} \right)$$

Find $\frac{\partial z}{\partial y}$

$$xz + x \ln y = z^2$$

$$x \cdot \frac{\partial z}{\partial y} + \frac{x}{y} = 2z \cdot \frac{\partial z}{\partial y}$$

$$\frac{x}{y} = 2z \cdot \frac{\partial z}{\partial y} - x \frac{\partial z}{\partial y}$$

$$\frac{x}{y} = \frac{\partial z}{\partial y} (2z - x)$$

$$\frac{\frac{x}{y}}{(2z - x)} = \frac{\partial z}{\partial y}$$

$$z + x \frac{\partial z}{\partial x} + \ln y = 2z \cdot \frac{\partial z}{\partial x}$$

$$x + \ln y = 2z \cdot \frac{\partial z}{\partial x} - x \frac{\partial z}{\partial x}$$

$$x + \ln y = \frac{\partial z}{\partial x} (2z - x)$$

$$\frac{\partial z}{\partial x} = \frac{x + \ln y}{2z - x}$$

- This problem has implicit differentiation in it. We were given an equation with multiple variables in it but we want it in terms of x but we must implicitly differentiate z as we go then isolate and solve for $\partial z / \partial x$ and we treat y as a constant.

$$\frac{\partial z}{\partial y} = \frac{x}{y(2z - x)}$$

$$f(x,y) = x^3 y^2 + 5xy - 7y$$

Find f_x and f_y

$$\frac{\partial f}{\partial x} = f_x = 3x^2 y^2 + 5y - 0$$

$$\frac{\partial f}{\partial y} = f_y = 2y x^3 + 5x - 7$$

Find f_{xx} , f_{xy} , f_{yx} , f_{yy}

$$\frac{\partial^2 f}{\partial x^2} = f_{xx} = 6x y^2$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = f_{xy} = 6x^2 y + 5$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = f_{yx} = 6x^2 y + 5$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = f_{yy} = 2x^3$$

The temperature at a point (x,y) on a flat metal plate is given by $T(x,y) = \frac{60}{1+x^2+y^2}$ where T is measured in $^{\circ}\text{C}$ and x,y in meters. Find the rate of change with respect to distance at point $(2,1)$

in x -direction

$$\frac{\partial T}{\partial x} @ (2,1) = \frac{60}{1+x^2+y^2} \Rightarrow 60(1+x^2+y^2)^{-1} = -120x(1+x^2+y^2)^{-2} = \frac{-120x}{(1+x^2+y^2)^2}$$

$$T_x(2,1) = \frac{-120(2)}{(1+(2)^2+(1)^2)^2} = \boxed{-\frac{20}{3}}$$

in y -direction

$$\frac{\partial T}{\partial y} @ (2,1) = \frac{60}{1+x^2+y^2} \Rightarrow 60(1+x^2+y^2)^{-1} \Rightarrow -120y(1+x^2+y^2)^{-2} = \frac{-120y}{(1+x^2+y^2)^2}$$

$$T_y(2,1) = \frac{-120(1)}{(1+(2)^2+(1)^2)^2} = \boxed{-\frac{10}{3}}$$

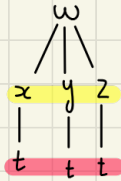
• Chain rule with Partial derivatives

ex:

$$w = \ln(x^2 + y^2 z^2) \text{ and } x = \sqrt[3]{t}, y = \sin t, z = \cos t$$

- these would require Partial derivatives because they have different Variable

$$\frac{\partial w}{\partial x}, \frac{\partial w}{\partial y}, \frac{\partial w}{\partial z}$$



- look at the equation for w there is a x, y and z in it if we take a look at each x, y and z they all have the Variable t

- these do not need Partial derivatives you

$$\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}$$

Just need to take the derivative because they all have the same Variable

$$\frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

$$\frac{\partial w}{\partial x} = \frac{2x}{x^2 + y^2 z^2}$$

★ Now after taking all the Partial derivatives and derivatives Just Plug them in

$$\frac{\partial w}{\partial y} = \frac{2y}{x^2 + y^2 z^2}$$

$$\frac{\partial w}{\partial z} = \frac{2zy^2}{x^2 + y^2 z^2}$$

$$\frac{\partial w}{\partial t} = -\frac{2x}{x^2 + y^2 z^2} \cdot \frac{1}{3} t^{-2/3} + \frac{2y}{x^2 + y^2 z^2} \cos t - \frac{2zy^2}{x^2 + y^2 z^2} (\sin t)$$

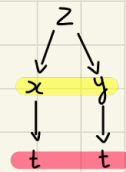
$$\frac{dx}{dt} = t^{-1/3} = -\frac{1}{3} t^{-4/3}$$

$$\frac{dy}{dt} = \cos t = \sin t$$

$$\frac{dz}{dt} = -\sin t = -\cos t$$

$$Z = xy^3 - x^2y, \quad x = t^2 + 1, \quad y = t^2 - 1$$

$$\frac{\partial Z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial Z}{\partial y} \cdot \frac{dy}{dt}$$



Partial derivatives

derivatives

$$\frac{\partial Z}{\partial x} = y^3 - 2xy$$

$$(y^3 - 2xy) \cdot 2t + (3y^2x - x^2) \cdot 2t$$

$$\frac{\partial Z}{\partial y} = 3y^2x - x^2$$

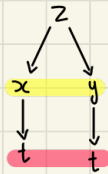
$$2ty^3 - 4xyt + 6ty^2x - 2tx^2$$

$$\frac{dx}{dt} = 2t$$

$$\frac{dy}{dt} = 2t$$

$$Z = \sin x \cos y, \quad x = \sqrt{t}, \quad y = \frac{1}{t}$$

$$\frac{\partial Z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial Z}{\partial y} \cdot \frac{dy}{dt}$$



Partial derivative

derivative

$$\frac{\partial Z}{\partial x} = \cos x \cos y$$

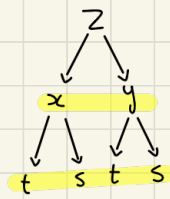
$$\frac{(\cos x \cos y)}{2\sqrt{t}} - \frac{(\sin x \sin y)}{t^2}$$

$$\frac{\partial Z}{\partial y} = -\sin x \sin y$$

$$\frac{dx}{dt} = t^{-1/2} = \frac{1}{2\sqrt{t}}$$

$$\frac{dy}{dt} = t^{-1} \Rightarrow -t^{-2}$$

$$z = x^2 e^{xy}, \quad x = \sqrt{ts}, \quad y = e^{t/s}$$



Partial derivatives

$$\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial x} = 2x e^{xy} + x^2 y e^{xy}$$

$$\frac{\partial z}{\partial y} = x^2 y e^{xy}$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx} \Rightarrow (2x)(e^{xy}) + (x^2)(y e^{xy})$$

$$f = x^2 \quad g = e^{xy}$$

$$f = x^2 \quad g = e^{xy}$$

$$\frac{df}{dy} = 0 \quad \frac{dg}{dy} = e^{xy} \cdot x$$

$$\frac{df}{dx} = 2x \quad \frac{dg}{dx} = e^{xy} \cdot y$$

$$(\cancel{0})(\cancel{e^{xy}}) + (x^2)(y e^{xy})$$

$$x^2 y e^{xy}$$

$$\frac{\partial x}{\partial t} = (ts)^{1/2} = \frac{1}{2} (ts)^{-1/2} \cdot s$$

$$= \frac{s}{2\sqrt{ts}}$$

$$\frac{\partial y}{\partial t} = e^{t/s} = \frac{e^{t/s}}{s}$$

$$\frac{\partial z}{\partial x} = (2x)(e^{xy}) + (x^2)(y e^{xy})$$

$$\frac{dF}{dx} \cdot g + F \cdot \frac{dg}{dx} \Rightarrow (2x)(e^{xy}) + (x^2)(y e^{xy})$$

$$F = x^2 \quad g = e^{xy}$$

$$\frac{dF}{dx} = 2x \quad \frac{dg}{dx} = y e^{xy}$$

$$\frac{\partial x}{\partial s} = (ts)^{1/2} = \frac{t}{2\sqrt{ts}}$$

$$\frac{\partial z}{\partial y} = x^3 e^{xy}$$

$$F = x^2 \quad g = e^{xy}$$

$$\frac{dF}{dy} = 0 \quad \frac{dg}{dy} = x e^{xy}$$

$$(\cancel{0})(\cancel{e^{xy}}) + (x^2)(x e^{xy})$$

$$\frac{\partial y}{\partial s} = e^{t/s} = -e^{t/s} \cdot \frac{t}{s^2}$$

$$\text{First equation: } 2x e^{xy} + x^2 y e^{xy} \cdot \left(\frac{s}{2\sqrt{ts}}\right) + x^2 y e^{xy} \cdot \left(\frac{e^{t/s}}{s}\right)$$

$$\text{Second equation: } (2x)(e^{xy}) + (x^2)(y e^{xy}) \cdot \left(\frac{t}{2\sqrt{ts}}\right) - x^3 e^{xy} \left(\frac{t e^{t/s}}{s^2}\right)$$

• implicit differentiation

$$\text{Find } dy/dx$$

$$y \cos x = x^2 + y^2$$

$$\frac{dF}{dx} \cdot g + F \cdot \frac{dg}{dx} \quad F = y \quad g = \cos x$$

$$\frac{dF}{dx} = \frac{dy}{dx}$$

$$\frac{dg}{dx} = -\sin x$$

$$\left(\frac{dy}{dx}\right)(\cos x) - (y)(\sin x)$$

$$\left(\frac{dy}{dx}\right)(\cos x) - (y)(\sin x) = 2x + 2y \cdot \frac{dy}{dx}$$

$$\left(\frac{dy}{dx}\right)(\cos x) - 2y \frac{dy}{dx} = 2x + y \sin x$$

$$\frac{dy}{dx} (\cos x - 2y) = 2x + y \sin x$$

$$\boxed{\frac{dy}{dx} = \frac{2x + y \sin x}{\cos x - 2y}}$$

- tangent Planes and linear approximations

equation of the tangent plane to the surface $z = f(x, y)$ at the point $P(x_0, y_0, z_0)$

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

linearization of the function $z = f(x, y)$ at the point $P(x_0, y_0, z_0)$ is given by

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

if $z = f(x, y)$ then the differential dz is also called the total differential is given by

$$dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$$

Find the equation of the tangent Plane to the given surface at the specific Points

a) $f(x,y) = x/y^2$ $(-4, 2, -1)$

$$Z - Z_0 = f_x(x - x_0) + f_y(y - y_0)$$

$\downarrow$ given $\downarrow$ given

• We are apply Partial derivatives

$$\frac{\partial f}{\partial x} @ (x_0, y_0, z_0)$$

$$\frac{\partial f}{\partial y} @ (x_0, y_0, z_0)$$

$$x_0 = -4, y_0 = 2, z_0 = -1$$

• Now Plug into the equation

$$f_x = \frac{1}{y^2} @ (4, 2, -1) = 1/4$$

$$Z - (-1) = 1/4 (x - (-4)) + y - 2$$

$$f_y = x \cdot y^{-2} = -2x/y^3 @ (4, 2, -1) = 1$$

$$Z + 1 = 1/4 (x + 4) + y - 2$$

$$F(x,y) = x^2 + xy + y^2$$

$$x_0 = 2, y_0 = 2, z_0 = 5$$

$$Z - Z_0 = f_x(x - x_0) + f_y(y - y_0)$$

$$Z - 5 = 6(x - 2) + 6(y - 2)$$

$$f_x = 2x + y @ (2, 2, 5) = 6$$

$$f_y = x + 2y @ (2, 2, 5) = 6$$

Find the linear approximations of the $f(x,y) = 1 - xy \cos(\pi y)$ at $(1,1)$ and use it to approximate the value of $f(1.02, 0.97)$

$$L(x,y) = f(x_0, y_0) + F_x(x-x_0) + F_y(y-y_0)$$

$$F_x = -y \cos(\pi y) = F(1,1) = -1 \cos(\pi) = 1$$

$$F_y = -x \cos(\pi y) - xy \pi \sin(\pi y) \Rightarrow F(1,1) = (-1) \cos(\pi) - \pi \sin(\pi) = 1$$

$$\frac{df}{dy} \cdot y + f \cdot \frac{dy}{dy} \quad (-x)(\cos(\pi y)) + (-xy)(-\pi \sin(\pi y))$$

$$F = -xy \quad g = \cos(\pi y)$$

$$\frac{df}{dy} = -x \quad \frac{dg}{dy} = -y \sin(\pi y)$$

$$F(x_0, y_0) = F(1,1) = 1 - (1)(1) \cos(\pi) = 2$$

(Plug into original)

Now Plug all the stuff you got into the $L(x,y) = F(x_0, y_0) + F_x(x-x_0) + F_y(y-y_0)$

$$L(x,y) = 2 + 1(x-1) + 1(y-1) \Rightarrow L(x,y) = x+y$$

$$\rightarrow f(1.02, 0.97) \approx L(1.02, 0.97) = 1.02 + 0.97 = \boxed{1.99}$$

Plug into $L(x,y)$

if $z = x^2 - xy + 3y^2$ and (x, y) change from $(3, -1)$ to $(2.96, -0.95)$, Compare the Value of Δz and dz

$\uparrow$
 Starting Point

$$L: dz = F_x dx + F_y dy$$

$dx = \text{change in } x = \text{end} - \text{start}$

$dy = \text{change in } y = \text{end} - \text{start}$

$$F_x = 2x - y = F_x(\text{start}) = 2(3) - (-1) = 7$$

$$F_y = -x + 6y = F_y(\text{start}) = -(3) + 6(-1) = -9$$

$$dx = 2.96 - 3 = 0.04$$

$$dy = -0.95 + 1 = 0.05$$

$$dz = 7(-0.04) - 9(0.05) = \boxed{-0.73}$$

$$\Delta z = F(\text{end}) - F(\text{start}) = F(2.96, -0.95) - F(3, -1) = 14.28 - 15 = \boxed{-0.72}$$

You are plug this into $z = x^2 - xy + 3y^2$ once for ending the another For starting then subtract them

the length and width of a rectangle are measured as 24 in and 50 in respectively with and error in measurement of at most 0.05 in each. Use differentials to estimate the maximum error in the calculated area of the rectangle



$$L = 24 \text{ inches}$$

$$\Delta L = 0.05 \text{ in}$$

$$w = 50 \text{ inches}$$

$$\Delta w = 0.05 \text{ in}$$

$$\text{area} = L \cdot w$$

$$\partial A = A_L \cdot dL + A_w \cdot dw$$

$$\frac{\partial A}{\partial L} = w$$

$$\frac{\partial A}{\partial w} = L$$

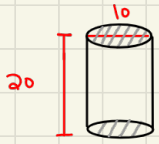
• Plug these into the equation

$$\partial A = w dL + L dw$$

$$(50)(0.05) + (24)(0.05)$$

$$\boxed{\partial A = 3.7 \text{ in}^2}$$

Use differential to estimate the amount of tin in the closed tin can with diameter 10cm and height 20 cm if the tin is 0.03 cm thick



$$\Delta r = 0.03$$

$$\Delta h = 0.03 + 0.03$$

★ top and bottom

$$V = \pi r^2 h \quad r = \frac{10}{2} = 5 \text{ cm}$$

$$dV = V_r \cdot dr + V_h \cdot dh$$

$$dV = 2\pi r h dr + \pi r^2 dh$$

$$dV = 2\pi (5)(20)(0.03) + \pi (5)^2 (0.06)$$

$$= \boxed{23.56 \text{ cm}^3}$$

The length l , width w , and height h of a box changes with time at a certain instant the dimensions are $l=1\text{m}$, $w=2\text{m}$, $h=3\text{m}$, l is increasing at a rate of 2m/s , w is increasing at a rate of 1m/s and h is increasing at a rate of 0.5m/s at that instant find the rate at which The volume, the surface area and length of the diagonal

$$\frac{dl}{dt} = 2 \text{ m/s}$$

$$l = 1 \text{ m}$$

$$\frac{dw}{dt} = 1 \text{ m/s}$$

$$w = 2 \text{ m}$$

$$h = 3 \text{ m}$$

$$\frac{dh}{dt} = 0.5 \text{ m/s}$$

$$V = L(t)w(t)h(t)$$

$$\frac{dV}{dt} = \frac{dL}{dt} \cdot w \cdot h + L \cdot \frac{dw}{dt} \cdot h + L \cdot w \cdot \frac{dh}{dt}$$

$$L = L(t)$$

$$w = w(t)$$

$$h = h(t)$$

$$\frac{dL}{dt} = \frac{dL}{dt}$$

$$\frac{dw}{dt} = \frac{dw}{dt}$$

$$\frac{dh}{dt} = \frac{dh}{dt}$$

$$\frac{dL}{dt} \cdot (w(t)) \cdot (h(t)) + (L(t)) \cdot \left(\frac{dw}{dt}\right) \cdot (h(t)) + (L(t)) \cdot (w(t)) \cdot \left(\frac{dh}{dt}\right)$$

$$(2) (2) (3) + (1) (1) (3) + (1) (2) (0.5)$$

$$\boxed{\text{Volume} = 16 \text{ m}^3/\text{s}}$$

Surface area

$$2w(t)h(t) + 2l(t)h(t) + 2l(t)w(t)$$

$$\begin{aligned} F &= 2w(t) & g &= h(t) \\ \frac{dF}{dx} \cdot g + F \cdot \frac{dg}{dh} & & \frac{dF}{dt} &= 2 \frac{dw}{dt} & \frac{dg}{dt} &= \frac{dh}{dt} \end{aligned}$$

$$\begin{aligned} F &= 2l(t) & g &= h(t) \\ \frac{dF}{dt} &= 2 \frac{dl}{dt} & \frac{dg}{dt} &= \frac{dh}{dt} \end{aligned}$$

$$2 \frac{dw}{dt} \cdot h(t) + 2w(t) \frac{dh}{dt}$$

$$2 \frac{dl}{dt} h(t) + 2l(t) \frac{dh}{dt}$$

$$\begin{aligned} F &= 2l(t) & g &= w(t) \\ \frac{dF}{dt} &= 2 \frac{dl}{dt} & \frac{dg}{dt} &= \frac{dw}{dt} \end{aligned}$$

$$2 \frac{dl}{dt} w(t) + 2l(t) \frac{dw}{dt}$$

$$2 \frac{dw}{dt} \cdot h(t) + 2w(t) \frac{dh}{dt} + 2 \frac{dl}{dt} h(t) + 2l(t) \frac{dh}{dt} + 2 \frac{dl}{dt} w(t) + 2l(t) \frac{dw}{dt}$$

$$2(1)(3) + 2(2)(0.5) + 2(2)(3) + 2(1)(3) + (2)(2)(2) + 2(1)(1)$$

$$= \boxed{34 \text{ m/s}}$$

• Maximum and minimum values

if $D > 0$ and $F_{xx}(a,b) > 0$ then $F(a,b)$ is a local minimum

if $D > 0$ and $F_{xx}(a,b) < 0$ then $F(a,b)$ is a local maximum

if $D < 0$ (a,b) is a Saddle Point

$$D = F_{xx}(a,b)F_{yy}(a,b) - F_{xy}(a,b)F_{yx}(a,b)$$

example #1:

$$F(x,y) = x^3 - 3x - 3xy^2$$

$$F_x = 3x^2 - 3 - 3y^2$$

$$F_y = -6xy$$

} • set equal to zero and solve the system of equations

✚ Set any equation to $x=0$ and $y=0$

when you set $x=0$

$$-3 - 3y^2 = 0$$

$$3y^2 = -3$$

$$\sqrt{y^2} = \sqrt{-1} = \pm i$$

the answer is

Not useful

ignore it

When you set $y=0$

$$3x^2 - 3 = 0$$

$$3x^2 = 3$$

$$\sqrt{x^2} = \sqrt{1} = \pm 1$$

You already set $y=0$ and got $x = \pm 1$ So the critical points are $(1,0)$ and $(-1,0)$

Find second derivative

$$F_{xx} = 6x$$

$$F_{xy} = -6y$$

$$F_{yy} = -6x$$

$$F_{yx} = -6y$$

• Plug these into

$$D = F_{xx}(a,b)F_{yy}(a,b) - F_{xy}(a,b)F_{yx}(a,b)$$

$$D = -36x^2 - 36y^2$$

@ $(1,0)$ $D = -36 < 0$ So saddle point at $(1,0)$

@ $(-1,0)$ $D = -36 < 0$ So another saddle point at $(-1,0)$